

Description of the Cleaned NHANES Dataset

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Introduction

The National Health and Nutrition Examination Survey (NHANES) is a nationally representative survey conducted by the Centers for Disease Control and Prevention (CDC) to assess the health and nutritional status of the population in the United States. NHANES combines interviews, physical examinations, and laboratory measurements to provide information about a wide range of health indicators (Centers for Disease Control and Prevention 2026).

This report describes the cleaned NHANES dataset used for analysis. The dataset combines demographic and blood pressure information from three NHANES waves, represented by the wave identifiers 2015, 2016, and 2017.

Sample Size and Biological Sex

The first table describes the overall sample size and distribution of male participants across the three NHANES waves.

Table 1: Sample Size and Male Participants by NHANES Wave

wave_id	Overall Sample	Male Participants	
	Sample Size	Male (n)	Male (%)
2015	10175	5003	49.2
2016	9971	4892	49.1
2017	9254	4557	49.2

Note:

Male percentages represent the percentage of participants identified as male within each wave.

The cleaned dataset contains substantial samples across all three waves, with 10,175 participants in 2015, 9,971 in 2016, and 9,254 in 2017. The male count and percentage provide a description of the biological sex distribution within each wave.

Ethnicity Distribution

The following table summarizes the distribution of ethnicity across the three NHANES waves. Counts and percentages are calculated within each wave.

Table 2: Ethnicity Distribution by NHANES Wave

Wave	Ethnicity	Participants	
		n	%
2015	Asian	1074	10.6
2015	Black	2267	22.3
2015	Mexican	1730	17.0
2015	Other	470	4.6
2015	Other Hispanic	960	9.4
2015	White	3674	36.1
2016	Asian	1042	10.5
2016	Black	2129	21.4
2016	Mexican	1921	19.3
2016	Other	505	5.1
2016	Other Hispanic	1308	13.1
2016	White	3066	30.7
2017	Asian	1168	12.6
2017	Black	2115	22.9
2017	Mexican	1367	14.8
2017	Other	634	6.9
2017	Other Hispanic	820	8.9
2017	White	3150	34.0

Note:

Percentages are calculated within each NHANES wave.

The ethnicity table demonstrates the diversity of the cleaned sample and allows the composition of participants to be compared across survey waves. The six ethnicity categories represented in the cleaned dataset are Black, White, Mexican, Other, Asian, and Other Hispanic.

Average Systolic Blood Pressure

Average systolic blood pressure was calculated from the available systolic blood pressure measurements for each participant. The following table summarizes the distribution of average systolic blood pressure within each NHANES wave.

Table 3: Summary Statistics for Average Systolic Blood Pressure by NHANES Wave

Wave	Average Systolic Blood Pressure (mm Hg)				
	Minimum	Maximum	Median	Mean	Standard Deviation
2015	64.7	228.7	115.3	118.3	18.1
2016	74.0	231.3	117.3	120.4	18.4
2017	72.7	234.0	118.0	121.7	20.3

Note:

Average systolic blood pressure was calculated from the available systolic blood pressure measurements for each participant. Values are reported in mm Hg.

The average systolic blood pressure summaries allow comparison of blood pressure distributions across the three NHANES waves. The mean and median provide measures of central tendency, while the minimum and maximum describe the observed range. The standard deviation provides information about the variability of average systolic blood pressure within each wave.

Conclusion

Overall, these tables provide a concise description of the cleaned NHANES dataset. The tables demonstrate how sample characteristics, ethnicity, and average systolic blood pressure can be summarized and compared across survey waves using descriptive statistics and formatted tables.

References

Centers for Disease Control and Prevention. 2026. “National Health and Nutrition Examination Survey.” National Center for Health Statistics. <https://www.cdc.gov/nchs/nhanes/>.